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2) Assignment #1:-

=> find & sketch the domain of the following function:-

(i) $f(x, y, z) = xy \ln z$

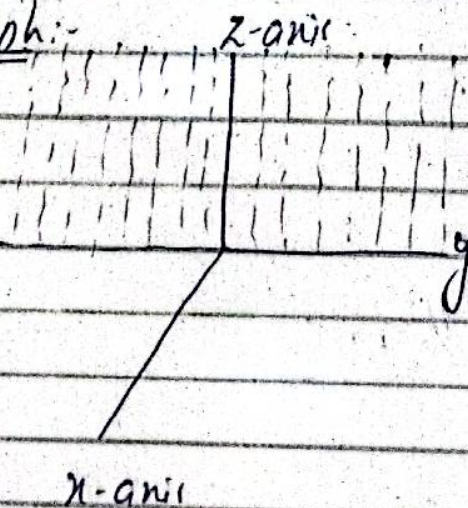
$f(x, y, z) = xy \ln z$

$\ln z \neq \infty$

Domain = $\{ (x, y, z) \mid z > 0 \}$

Ans

Graph:-



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(ii) $f(x,y) = \ln(x^2+y^2)$
soln

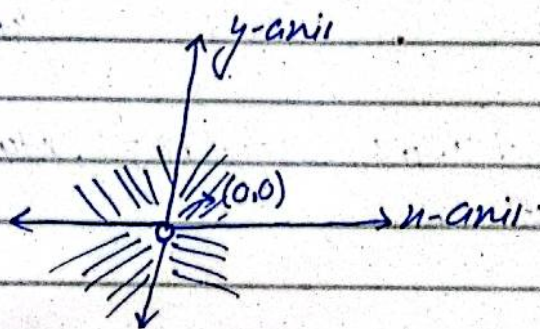
$$f(x,y) = \ln(x^2+y^2)$$

$\ln \neq \infty$

$$x^2+y^2 \neq 0$$

$$\text{Domain} = \{(x,y) / x^2+y^2 \neq 0\}$$

Graph:-



(iii) $f(x,y) = \sin^{-1}(y-x)$
soln

$$f(x,y) = \sin^{-1}(y-x)$$

$$\text{Domain} = \sin^{-1}(y-x) \Rightarrow -1 \leq y-x \leq 1$$

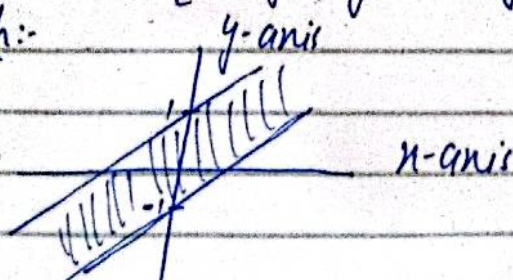
$$-1 \leq y-x \leq 1$$

$$y \geq x-1$$

$$y \leq x+1$$

$$\text{Domain} = \{(x,y) / y \geq x-1 / y \leq x+1\}$$

Graph:-



$$(iv) f(x, y, z) = \frac{1}{x+1} + \frac{1}{y-1} + \frac{1}{x+y-z}$$

Sol.

$$f(x, y, z) = \frac{1}{x+1} + \frac{1}{y-1} + \frac{1}{x+y-z}$$

$$x+1 \neq 0$$

$$x \neq -1$$

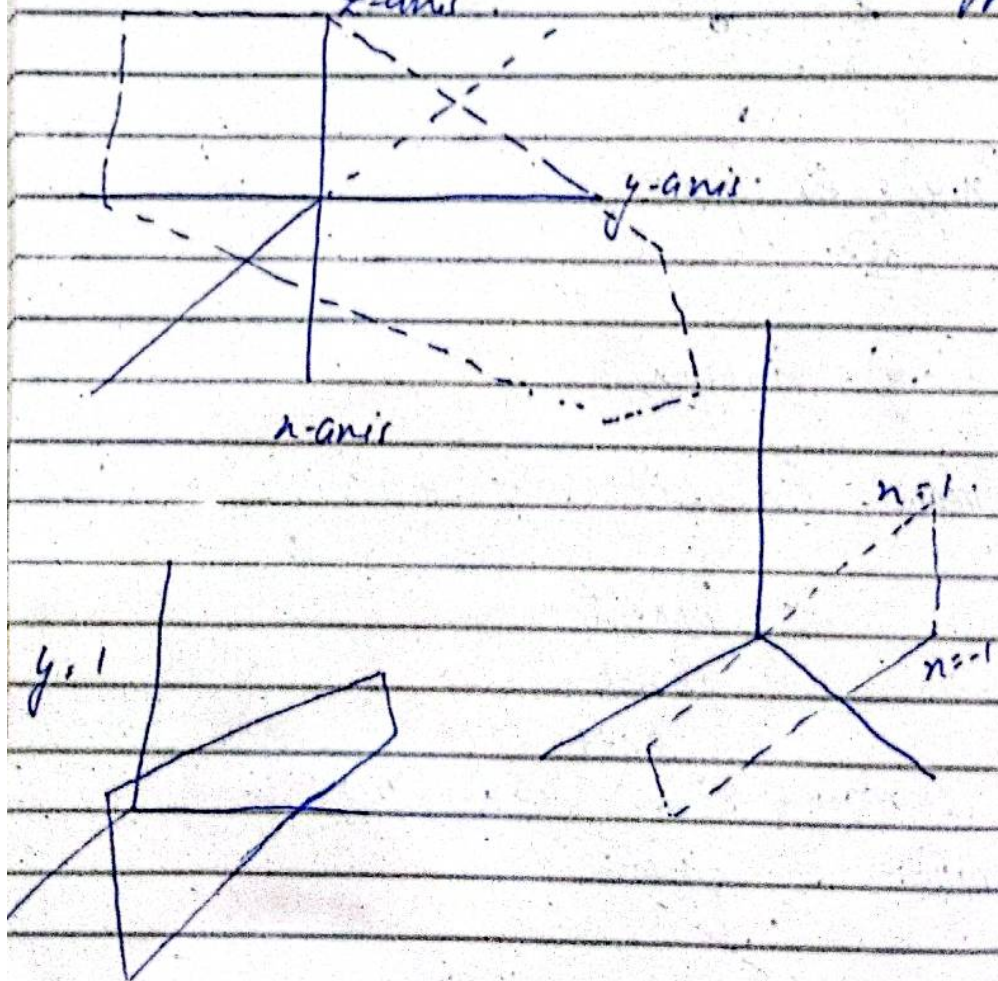
$$y-1 \neq 0$$

$$y \neq 1$$

$$x+y-z \neq 0$$

$$x+y \neq z$$

Domain: $\{(x, y, z) / x \neq -1, y \neq 1, x+y \neq z\}$



⇒ Draw an assortment of level curves in the function's domain.

1. $f(x,y) = 4x^2 + y^2 + 1$

for

$f(x,y) = 4x^2 + y^2 + 1 \quad k = 2, 3, 4, 5.$

$(x,y) = z = k.$

$k = 4x^2 + y^2 + 1.$

$4x^2 + y^2 + 1 = 2.$

$4x^2 + y^2 = 2 - 1$

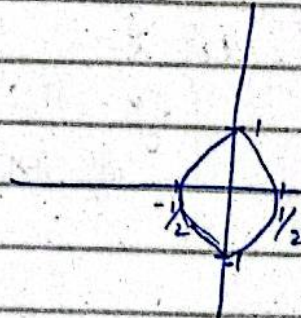
$4x^2 + y^2 = 1$

⇒ $\frac{x^2}{1/4} + \frac{y^2}{1} = 1$

$4x^2 + 0^2 = 1$

$x^2 = \frac{1}{4}$

$x = \pm \frac{1}{2}$



$k = 3$

$4x^2 + y^2 + 1 = 3$

$4x^2 + y^2 = 3 - 1$

$4x^2 + y^2 = 2$

$\frac{x^2}{1/2} + \frac{y^2}{2} = 1$

$\frac{4x^2}{2} + \frac{y^2}{2} = 1$

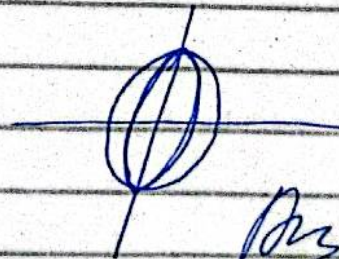
$\frac{x^2}{1/2} + \frac{y^2}{2} = 1$

$a^2 = \frac{1}{2}$

$b^2 = 2$

$a = \frac{1}{\sqrt{2}}$

$b = \sqrt{2}$



⇒ family of Ellipsoids.

$$2. \quad -2 = 2x - 6y + z.$$

for

$$-2 = 2x - 6y + z.$$

$$k = 0, 1, 2, 3$$

$$z = -2x + 6y - 2$$

$$z = k.$$

$$-2x + 6y - 2 = k.$$

$$-2x + 6y - 2 = 0.$$

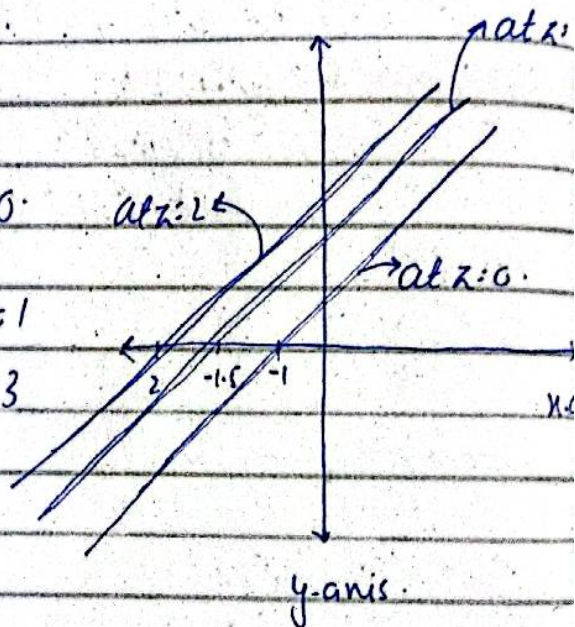
$$2(3y - x - 1) = 0.$$

$$3y - x - 1 = 0.$$

$$\text{at } x = 1$$

$$6y - 2x - 2 = 1$$

$$6y - 2x = 3$$



Q=3:- Sketch the level surface for the given function at the specific value of k .

$$1. \frac{x^2}{25} + \frac{y^2}{16} + \frac{z^2}{9} = k$$

Sol.

$$\frac{x^2}{25} + \frac{y^2}{16} + \frac{z^2}{9} = k$$

xy-plane:-
 $z=0$

$$\frac{x^2}{25} + \frac{y^2}{16} = k$$

$$y = \pm 4$$

$$x = \pm 5$$

xz-plane:-

$$y=0$$

$$\frac{x^2}{25} + \frac{z^2}{9} = k$$

$$y = \pm 5$$

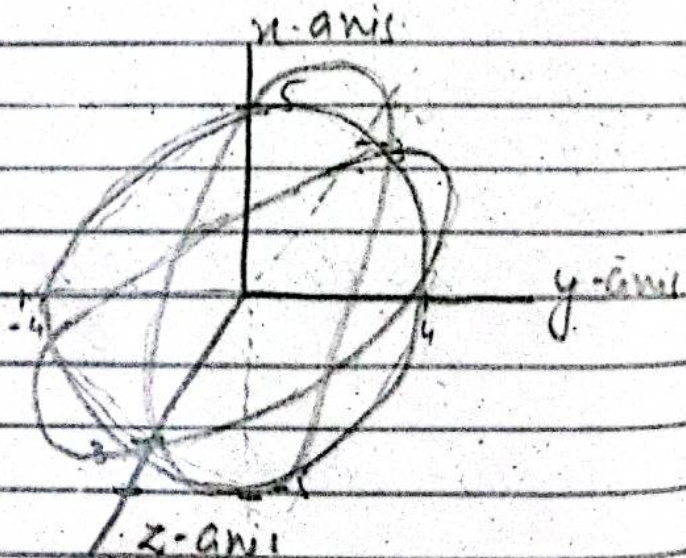
$$x = \pm 5$$

$$z = \pm 3$$

yz-plane:-

$$x=0$$

$$\frac{y^2}{16} + \frac{z^2}{9} = k$$



$$2. f(x, y, z) = 9x^2 + 4y^2 + z^2, k=4.$$

$$f(x, y, z) = 9x^2 + 4y^2 + z^2$$

$$w = 9x^2 + 4y^2 + z^2$$

$$w = k = 4$$

$$9x^2 + 4y^2 + z^2 = 4$$

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1} \rightarrow \text{Ellipsoid.}$$

$$\frac{9x^2}{4} + \frac{4y^2}{4} + \frac{z^2}{4} = \frac{4}{4} \cdot 1$$

$$\frac{x^2}{\frac{4}{9}} + y^2 + \frac{z^2}{4} = 1$$

xy -plane
 $z=0$

$$\frac{x^2}{\frac{4}{9}} + y^2 = 1$$

$$y = \pm 1$$

$$x = \pm \frac{2}{3}$$

xz -plane
 $y=0$

$$\frac{x^2}{\frac{4}{9}} + \frac{z^2}{4} = 1$$

$$x = \pm \frac{2}{3}$$

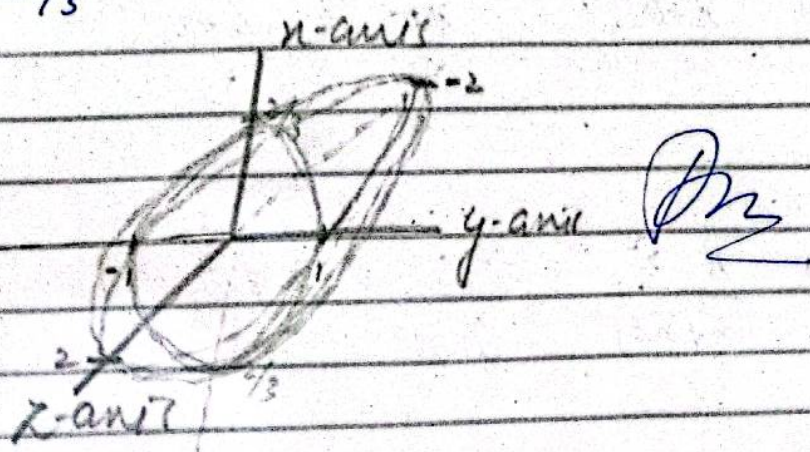
$$z = \pm 2$$

yz -plane
 $x=0$

$$y^2 + \frac{z^2}{4} = 1$$

$$y = \pm 1$$

$$z = \pm 2$$



Q#4:

find:

$$1. \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}} = \frac{(0)^2 - (0)(0)}{\sqrt{0} - \sqrt{0}} = \frac{0}{0}$$

x-axis

y=0

$$\lim_{x \rightarrow 0} \frac{x^2 - x(0)}{\sqrt{x} - \sqrt{0}} \Rightarrow \frac{x^2}{\sqrt{x}} = \frac{0}{0}$$

y-axis

x=0

$$\lim_{y \rightarrow 0} \frac{(0)^2 - (0)y}{\sqrt{0} - \sqrt{y}} \Rightarrow \frac{-0}{\sqrt{y}} \Rightarrow 0$$

y=x

$$\lim_{y=x} \frac{x^2 - x(x)}{\sqrt{x} - \sqrt{x}} \Rightarrow \frac{x^2 - x^2}{0} = \frac{0}{0}$$

$\Rightarrow$ limit doesn't exist.

Q#4:-

$$2. \lim_{(x,y) \rightarrow (0,0)} \frac{\cos x^2 - y^2}{x + y + 1}$$

Sol:-

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\cos x^2 - y^2}{x + y + 1} \Rightarrow \frac{\cos(0)^2 - (0)^2}{0 + 0 + 1}$$

$$\frac{\cos(0)^2 - (0)^2}{0 + 0 + 1} \Rightarrow \cos(0) \Rightarrow \cos 0 = 1.$$

y-axis
x=0

$$\lim_{y \rightarrow 0} \frac{x^2 - y^2}{x + y + 1} \Rightarrow \frac{\cos(0)^2 - y^2}{0 + y + 1} \Rightarrow \frac{-y^2}{y + 1} \Rightarrow \frac{0^2}{0 + 1} \Rightarrow \frac{0}{1} = 0$$

$\cos 0 = 1$

x-axis

y=0

$$\lim_{x \rightarrow 0} \frac{x^2 - (0)^2}{x + (0) + 1} \Rightarrow \frac{\cos x^2}{x + 1} \Rightarrow \frac{\cos(0)^2}{0 + 1} \Rightarrow \frac{0}{1} = \cos(0)$$

$= 1$

Ans

y=x

$$\lim_{y \rightarrow x} \frac{x^2 - y^2}{x + y + 1} \Rightarrow \frac{0}{2x + 1} = \frac{0}{1} = 0$$

Q#5:-

$$V = \frac{1}{3} \pi r^2 h$$

$$\frac{dr}{dt} = -2 \text{ mm}$$

$$\frac{dh}{dt} = 3 \text{ mm}$$

Sol:-

$$\frac{dV}{dt} = \frac{1}{3} \frac{dV}{dr} \frac{dr}{dt} + \frac{dV}{dh} \frac{dh}{dt}$$

$$\frac{dV}{dt} = \left[\frac{2}{3} \pi r h \cdot -0.2 \text{ cm} \right] + \left[\frac{1}{3} \pi r^2 \cdot 0.3 \text{ cm} \right]$$

$$= \left[\frac{2(3.14)(1.5)(3.2)(-0.2 \text{ cm})}{3} \right] + \left[\frac{1(3.14)(1.5)^2(0.3)}{3} \right] \quad r = 1.5 \text{ cm}$$

$$h = 3.2 \text{ cm}$$

$$= [-0.628] +$$

$$= \frac{[-2.0096] + [3.132]}{3} = [-1.3031 \text{ cm}^3/\text{s}] \text{ Ans.}$$

Q=) 6..

$$A = \frac{1}{2} ac \sin B$$

$$\frac{da}{dt} = 0.4$$

Sol.

$$\frac{dc}{dt} = -0.8$$

$$\frac{dA}{dt} = \frac{dA}{da} \cdot \frac{da}{dt} + \frac{dA}{dc} \cdot \frac{dc}{dt} + \frac{dA}{dB} \cdot \frac{dB}{dt}$$

$$\frac{dB}{dt} = 0.2$$

$$A = \frac{1}{2} ac \sin A$$

$$\frac{dA}{da} = \frac{1}{2} c \sin A$$

$$\frac{dA}{dc} = \frac{1}{2} a \sin A$$

$$\frac{dA}{dB} = \frac{1}{2} ac \cos B \cdot 1 \Rightarrow \frac{1}{2} ac \cos A$$

$$\frac{dA}{dt} = \left[\frac{1}{2} c \sin A \cdot 0.4 \right] + \left[\frac{1}{2} a \sin A \cdot (-0.8) \right] + \left[\frac{1}{2} ac \cos A \cdot 0.2 \right]$$

$$a = 3$$

$$c = 4$$

$$\left[\frac{1}{2} (4) \sin(30^\circ) \cdot 0.4 \right] + \left[\frac{1}{2} (3) \sin(30^\circ) \cdot (-0.8) \right] + \left[\frac{1}{2} (3)(4) \cos(30^\circ) \cdot 0.2 \right]$$

$$= \left[\frac{1 \cdot 4 \cdot 1 \cdot 0.4}{2 \cdot 1} \right] + \left[\frac{1 \cdot 3 \cdot 0.5 \cdot (-0.8)}{2 \cdot 1} \right] + \left[\frac{6 \cdot 0.866}{2} \cdot 0.2 \right]$$

$$= [0.4] + [-0.6] + [1.14312]$$

$$= 0.94312 \text{ Ans.}$$

Q47:

$$PV^{1.4} = k.$$

~~Q47~~

$$PV^{1.4} = k.$$

Apply $\ln$:

$$\frac{dP}{dt} = \frac{4}{100}$$

$$\ln(PV^{1.4}) = \ln k$$

$$\frac{dV}{dt} = \frac{-1.5}{100}$$

$$\ln P + \ln V^{1.4} = \ln k$$

$$\frac{1}{P} \cdot \frac{dP}{dt} + \frac{1.4}{V} \cdot \frac{-1.5}{100} = 0.$$

$$\frac{4V - 2.1P}{PV100} = 0.$$

$$4V = 2.1P$$

$$\frac{P}{V} = \frac{2.1}{2.1} = 1.9\% \text{ Error } \text{Ans.}$$

Q#8:

$$w: f(T, v)$$

$$\frac{\delta w}{\delta x}$$

$$f_T = (-15, 30)$$
$$f_v = (-15, 30)$$

Using

$$f_T \Rightarrow \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\Delta x = -5$$

$$= \frac{f(-15 + (-5), 30) - f(-15, 30)}{-5}$$

$$= \frac{f(-20, 30) - f(-15, 30)}{-5}$$

$$= \frac{-33 - (-26)}{-5} \Rightarrow \frac{-7}{-5} \Rightarrow 1.4 \text{ Ans}$$

$$f_v \Rightarrow \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \quad \Delta y = 10$$

$$\Rightarrow \frac{f(-15, 30 + 10) - f(-15, 30)}{10}$$

$$\Rightarrow \frac{-27 - (-26)}{10}$$

$$\Rightarrow \frac{-1}{10} = -0.1 \text{ Ans}$$

Q#9:

(a) $\frac{\partial h}{\partial v}$ (Partial derivative of h w.r.t to v):- This

represents the rate of change of h with respect to v while keeping all other variables constant.

→ It tells us how h changes when v changes, assuming that ~~to~~ any other variable doesn't vary.

$\frac{\partial h}{\partial t}$ (Partial derivative of h w.r.t to t):- This

represents the rate of change of h with respect to t while keeping all other variables constant.

→ It tells us how h changes over time, assuming that v or any other variable doesn't change.

$$(b) f_v(40, 15) \quad f_t(40, 15) \quad \hookrightarrow f(v, t)$$

$$\frac{f(x+\Delta x, y) - f(x, y)}{\Delta x} \quad \Delta x = 10$$

$$\frac{f(40+10, 15) - f(40, 15)}{10} \Rightarrow \frac{36-25}{10} = \frac{11}{10} = 1.1 \quad \text{Ans}$$

$$\lim_{\Delta t \rightarrow 0} \frac{f(v, t+\Delta t) - f(v, t)}{\Delta t}$$

$$\frac{f(40, 20) - f(40, 15)}{5} \Rightarrow \frac{28-25}{5} = \frac{3}{5} = 0.6 \quad \text{Ans}$$

Q=10:-

Directional Derivative:-

$$(1) f(x, y, z) = xe^y + ye^z + ze^x \quad (0, 0, 0) \\ v = (5, 1, -2) \\ \text{Soln}$$

$$\frac{df}{dx} = e^y + ze^x$$

$$\frac{df}{dy} = xe^y + e^z$$

$$\frac{df}{dz} = ye^z + e^x$$

$$\nabla f(0, 0, 0) = (e^0 + 0e^0, 0e^0 + e^0, 0e^0 + e^0) = (1, 1, 1)$$

$$|v| = \sqrt{(5)^2 + (1)^2 + (-2)^2}$$

$$= \sqrt{25 + 1 + 4}$$

$$= \sqrt{30}$$

$$u = \frac{1}{\sqrt{30}} (5, 1, -2)$$

$$= \left(\frac{5}{\sqrt{30}}, \frac{1}{\sqrt{30}}, \frac{-2}{\sqrt{30}} \right) \text{ Ans.}$$

$$= \frac{5}{\sqrt{30}} + \frac{1}{\sqrt{30}} - \frac{2}{\sqrt{30}} \Rightarrow \frac{4}{\sqrt{30}} \text{ Ans.}$$

(b) $f(x,y,z) = \sqrt{xyz}$ at $(3,2,6)$ vs $(-1,-2,2)$

Sol:-

$$\frac{1}{2} (xyz)^{-1/2} yz = \frac{yz}{2\sqrt{xyz}}$$

$$\frac{df}{dy} = \frac{xz}{2\sqrt{xyz}}$$

$$\frac{df}{dz} = \frac{xy}{2\sqrt{xyz}}$$

$$\nabla f(3,2,6) = \left(\frac{2(6)}{2\sqrt{3(2)(6)}}, \frac{3(6)}{2\sqrt{3(2)(6)}}, \frac{3(2)}{2\sqrt{3(2)(6)}} \right)$$

$$\left(\frac{12}{2\sqrt{36}}, \frac{18}{2\sqrt{36}}, \frac{6}{2\sqrt{36}} \right)$$

$$\left(\frac{12}{12}, \frac{18}{12}, \frac{6}{12} \right) = (1, 1.5, 0.5)$$

$$|\nabla f| = \sqrt{(-1)^2 + (-2)^2 + (2)^2} = \sqrt{1+4+4} = \sqrt{9} = 3$$

$$u = \frac{1}{3} (-1, -2, 2)$$

$$Df \cdot 4 = (1, 1.5, 0.5) \cdot \left(-\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right)$$

$$(1)\left(-\frac{1}{3}\right) + (1.5)\left(-\frac{2}{3}\right) + (0.5)\left(\frac{2}{3}\right)$$

$$= \frac{-1}{3} - \frac{3}{3} + \frac{1}{3}$$

$$= \frac{-3}{3} = -1 \quad \text{Ans.}$$